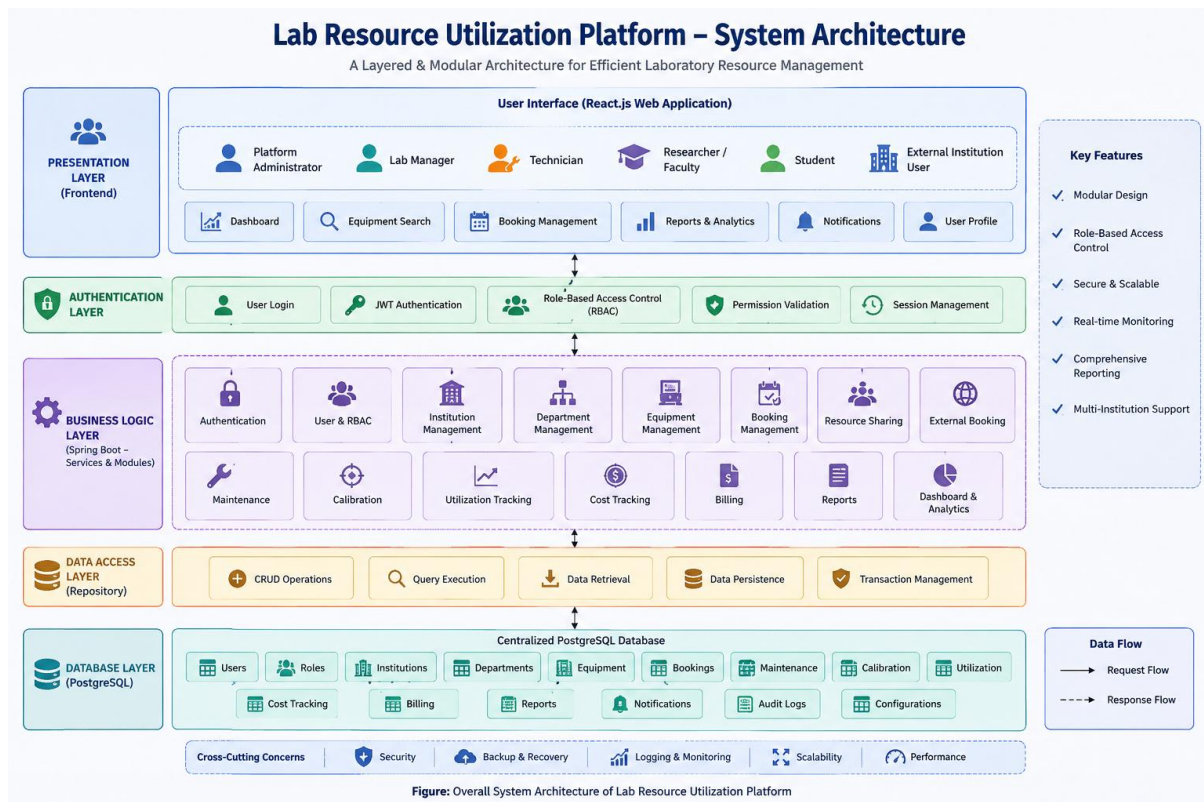


System Architecture

6.1 Introduction

The **Lab Resource Utilization Platform** is designed using a **layered and modular architecture** to ensure scalability, maintainability, security, and efficient management of laboratory resources. The system separates the application into multiple layers, where each layer has a specific responsibility. This architecture enables seamless communication between the user interface, business logic, and the database while maintaining a clean separation of concerns.

6.2 Overall System Architecture



The architecture consists of five major layers:

- Presentation Layer
- Authentication Layer
- Business Logic Layer
- Data Access Layer
- Database Layer

Each layer communicates with the adjacent layer through REST APIs and Spring Boot services to provide a secure and efficient workflow.

6.3 Presentation Layer (Frontend)

The Presentation Layer provides the graphical user interface through which users interact with the system. It is developed using **React.js** and offers responsive web pages for different user roles.

Responsibilities

- User Login
- Dashboard
- Equipment Search
- Booking Management
- Reports & Analytics
- Notifications
- User Profile Management

Supported Users

- Platform Administrator
- Lab Manager
- Technician
- Researcher / Faculty
- Student
- External Institution User

Technology Used: React.js, HTML, CSS, JavaScript

6.4 Authentication Layer

The Authentication Layer is responsible for validating users and controlling secure access to the application.

Responsibilities

- User Login Authentication

- JWT Token Generation
- Role Verification
- Permission Validation
- Session Management

Security Features

- JWT Authentication
 - Role-Based Access Control (RBAC)
 - Secure REST API Communication
-

6.5 Business Logic Layer

The Business Logic Layer is the core of the application and is implemented using **Spring Boot**. It contains all business rules and service classes responsible for processing requests and managing laboratory resources.

Major Modules

- Authentication
- User & RBAC Management
- Institution Management
- Department Management
- Equipment Management
- Booking Management
- Resource Sharing
- External Booking
- Maintenance Management
- Calibration Management
- Utilization Tracking
- Cost Tracking
- Billing Management
- Reports
- Dashboard & Analytics

Each module performs specific operations and communicates with other modules whenever required.

6.6 Data Access Layer

The Data Access Layer provides communication between the business logic and the PostgreSQL database. It is implemented using **Spring Data JPA Repositories**.

Responsibilities

- CRUD Operations
- Query Execution
- Data Retrieval
- Data Mapping
- Transaction Management
- Data Persistence

This layer ensures efficient and reliable interaction with the database.

6.7 Database Layer

The Database Layer stores all application data in a centralized **PostgreSQL Database**.

Database Entities

- Users
- Roles
- Institutions
- Departments
- Equipment
- Bookings
- Maintenance Records
- Calibration Records
- Utilization Records
- Cost Tracking
- Billing Records

- Resource Sharing
- External Bookings
- Reports
- Notifications
- Audit Logs

The centralized database maintains data integrity and supports efficient data retrieval across all modules.

6.8 Cross-Cutting Concerns

The architecture incorporates several cross-cutting services that improve the overall quality of the application.

Features

- Security
- Logging
- Monitoring
- Backup & Recovery
- Error Handling
- Scalability
- Performance Optimization

These services operate across all layers to ensure system reliability and availability.

6.9 Data Flow

The application follows a layered request-response flow.

1. User sends a request through the React application.
2. The Authentication Layer validates the user.
3. The request is forwarded to the appropriate Controller.
4. The Service Layer executes business logic.
5. The Repository Layer communicates with PostgreSQL.
6. Data is retrieved or updated.

7. The response is returned to the user through the frontend.

6.10 Key Features of the Architecture

- Layered Architecture
 - Modular Design
 - Secure JWT Authentication
 - Role-Based Access Control (RBAC)
 - REST API Communication
 - Centralized PostgreSQL Database
 - Real-time Monitoring
 - Multi-Institution Support
 - Comprehensive Reporting
 - Dashboard & Analytics
 - Scalable and Maintainable Design
-

6.11 Technologies Used

Layer	Technology
Frontend	React.js
Backend	Spring Boot
Authentication	JWT
Authorization	RBAC
Database	PostgreSQL
ORM	Spring Data JPA
API	REST API
Build Tool	Maven
Version Control	Git & GitHub

6.12 Advantages

- Improves code maintainability.
 - Supports modular development.
 - Provides secure authentication and authorization.
 - Enables easy integration of new modules.
 - Ensures centralized data management.
 - Supports multiple institutions and laboratories.
 - Enhances system scalability and performance.
-

6.13 Conclusion

The layered architecture of the Lab Resource Utilization Platform provides a structured approach for developing a secure, scalable, and maintainable laboratory management system. By separating the presentation, authentication, business logic, data access, and database layers, the system ensures efficient communication between components while supporting future enhancements and seamless integration of additional modules.
